

SkillSense AI - Architecture Overview

1. Introduction

SkillSense AI is an intelligent system designed to analyze a candidate's resume against a job description and provide actionable insights such as matched skills, skill gaps, interview questions, and a personalized learning plan.

2. System Components

The system consists of four primary components: • Frontend (Next.js/React): Handles user input and displays results. • Backend (FastAPI): Processes requests and orchestrates logic. • Processing Layer: Extracts, cleans, and matches skills. • AI Model (OpenAI/OpenRouter): Enhances extraction and generates outputs.

3. Frontend Layer

The frontend is built using Next.js and React. It provides interfaces for users to paste text or upload PDFs. It communicates with the backend using HTTP requests via Axios and renders results such as matched skills and skill gaps.

4. Backend Layer

The backend is implemented using FastAPI. It exposes endpoints such as /analyze, /analyze-pdf, /ask, /evaluate, and /learning-plan. It handles input validation, API calls, and response formatting.

5. Processing Layer

This layer performs key operations including: • PDF text extraction using PDFPlumber • Skill extraction using AI and fallback keyword matching • Skill normalization and cleaning • Matching logic using set operations to determine overlaps and gaps

6. AI Integration

The system integrates with an AI model (OpenAI/OpenRouter) to: • Extract structured skills from unstructured text • Generate interview questions • Evaluate answers • Create personalized learning plans

7. Data Flow

1. User inputs resume and job description 2. Frontend sends data to backend 3. Backend processes and calls AI model 4. Skills are extracted and matched 5. Results are returned to frontend 6. UI displays insights to the user

8. Conclusion

SkillSense AI provides a scalable and modular architecture that combines traditional processing with AI capabilities to deliver meaningful career insights.